Entropy-SGD Optimizes the Prior of a PAC-Bayes Bound

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Abstract

This report is an overview of the paper of Dziugaite and Roy : *Entropy-SGD Optimizes the Prior of a PAC-Bayes Bound: Generalization properties of Entropy-SGD and data-dependent priors [3].*

1 Introduction

Optimization remains a central pillar of machine learning, yet the mechanisms by which specific optimization algorithms achieve generalization are not fully understood. While standard Stochastic Gradient Descent performs well in many overparametrized settings, it is also prone to rapid overfitting when exposed to noisy data or random labels. To address this, algorithms such as Entropy-SGD have been proposed to bias the optimization process toward flat minima-regions of the loss landscape where the empirical risk is insensitive to small perturbations, which is empirically linked to superior generalization.

The work of Dziugaite and Roy provides a rigorous theoretical foundation for this heuristic by connecting local entropy optimization to PAC-Bayesian framework. They demonstrate that Entropy-SGD essentially optimizes the prior of a PAC-Bayes bound, a process that traditionally violates the requirement that the prior be data-independent. To resolve this, the authors introduce Entropy-SGLD, replacing the deterministic updates with Stochastic Gradient Langevin Dynamics. This transition allows the use of Differential Privacy to justify data-dependent priors, thereby providing valid generalization certificates.

In this report, we provide an overview of the mathematical equivalence between local entropy and PAC-Bayes bounds. We further detail our implementation of the Entropy-SGLD algorithm, discussing the significant computational challenges encountered during parameter tuning and the resulting trade-offs between training efficiency and the tightness of theoretical bounds. To highlight this, we present empirical results across three datasets : binary MNIST, random-label MNIST, and FashionMNIST. We conclude with a comprehensive analysis of the strengths and weaknesses of both the article's theoretical approach and the practical utility of the Entropy-SGLD method

2 Mathematical Framework

2.1 Entropy-SGD and Entropy-SGLD

Deep neural networks possess highly non-convex loss landscapes. To discover minima that generalize well, Chaudhari et al. [1] proposed **Entropy-SGD**, an algorithm designed to bias gradient descent toward wide valleys (flat minima), which empirically yield better generalization than sharp minima [4].

It achieves this by optimizing a smoothed loss called *Local Entropy* via two nested loops: an inner loop computing a local moving average μ, and an outer loop pulling the main weights w toward μ.

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While highly effective, its deterministic formulation challenges theoretical guarantees. To bridge this gap, the paper introduces **Entropy-SGLD** (Algorithm 1). By injecting specific Gaussian noise into the outer loop (Line 11), the algorithm becomes randomized, providing Differential Privacy (DP) properties that are fundamental to unlocking valid PAC-Bayesian bounds.

Algorithm 1 One outerloop step of Entropy-SG(L)D

Require: $w \in \mathbb{R}^p$ ▷ Current weights
$S \in \mathcal{Z}^m$ ▷ Data
$l : \mathbb{R}^p \times \mathcal{Z} \to \mathbb{R}$ ▷ Loss
$\tau, \beta, \gamma, \eta, \eta', L, K$ ▷ Parameters
Ensure: Weights w moved along stochastic gradient

1: **procedure** ENTROPY-SG(L)D-STEP
2: $w' \leftarrow w$
3: $\mu \leftarrow w$
4: **for** $i \in \{1, \dots, L\}$ **do** ▷ Run SG(L)D for L iterations.
5: $\eta'_i \leftarrow \eta'/i$
6: $(z_{j_1}, \dots, z_{j_K}) \leftarrow$ sample minibatch of size K
7: $dw' \leftarrow -\frac{1}{K}\sum_{k=1}^{K} \nabla_{w'} l(w', z_{j_k}) - \gamma(w' - w)$
8: $w' \leftarrow w' + \frac{1}{2}\eta'_i dw' + \sqrt{\eta'_i}\mathcal{N}(0, I_p)$
9: $\mu \leftarrow (1-\alpha)\mu + \alpha w'$
10: **end for**
11: $w \leftarrow w + \frac{1}{2}\eta\tau\gamma(\mu - w) + \underbrace{\sqrt{\eta/\beta}\mathcal{N}(0, I_p)}_{\text{Entropy-SGLD only}}$
12: **return** w
13: **end procedure**

2.2 PAC-Bayesian Theorem

The PAC-Bayesian framework [5] bounds the expected true risk $R(Q)$ of a randomized classifier (posterior Q) using its empirical training risk $\hat{R}_S(Q)$ and its Kullback-Leibler (KL) divergence from a prior P, which classically must be data-independent. For a confidence level $\delta \in (0, 1)$ and a dataset of size n, the classic McAllester bound is:

$$R(Q) \leq \hat{R}_S(Q) + \sqrt{\frac{\mathrm{KL}(Q \,||\, P) + \ln\left(\frac{2\sqrt{n}}{\delta}\right)}{2n}} \tag{1}$$

This highlights a fundamental trade-off: a model must achieve low empirical error while maintaining a low KL divergence between Q and P. However, applying this theory to Entropy-SGD creates a *data-dependent prior*, because the algorithm optimizes its prior based on the empirical risk landscape, which invalidates the classical PAC-Bayes bound.

2.3 Data-Dependent Priors and Differential Privacy

To reconcile empirical success with statistical validity, the authors of the paper, Dziugaite and Roy, rely on Differential Privacy (DP).

Definition 1 (ϵ-Differential Privacy [2])**.** *A randomized algorithm $\mathcal{A} : \mathcal{Z}^m \to \mathcal{T}$ is ϵ-differentially private if for all datasets $S, S' \in \mathcal{Z}^m$ differing by at most one element, and all measurable subsets of outputs $O \subseteq \mathcal{T}$:*

$$\mathbb{P}[\mathcal{A}(S) \in O] \leq e^{\epsilon}\mathbb{P}[\mathcal{A}(S') \in O] \tag{2}$$

In machine learning, this ensures the learned weights do not change significantly if a single training example is modified. An ϵ-DP data-dependent prior yields a valid, though slightly looser, PAC-Bayes bound:

Theorem 1 (DP PAC-Bayes Bound [3]). *Under* 0–1 *loss, for every* $\delta \in (0,1)$, $m \in \mathbb{N}$, *data distribution* $\mathcal{D}$, *and* ϵ*-differentially private data-dependent prior* $P : \mathcal{Z}^m \to \mathcal{M}_1(\mathbb{R}^p)$, *with probability at least* $1-\delta$ *over the draw of* $S \sim \mathcal{D}^m$, *the following holds for every posterior* $Q \in \mathcal{M}_1(\mathbb{R}^p)$*:*

$$KL(\hat{R}_S(Q) \| R_{\mathcal{D}}(Q)) \leq \frac{1}{m}\left(KL(Q \,\|\, P(S)) + \ln\frac{2\sqrt{m}}{\delta} + 2\max\left\{\ln\frac{3}{\delta}, m\epsilon^2\right\}\right) \tag{3}$$

Because sampling exactly from the local entropy distribution to achieve this privacy is computationally intractable, **Entropy-SGLD** injects Langevin noise into the outer loop (Algorithm 1, Line 11). This makes the sampling $\frac{2\beta L_{max}\tau}{m}$-differentially private, with L_{max} the length of the loss value interval. The parameter β regulates this privacy level, limiting the information the prior leaks from the training data while maintaining a mathematically rigorous bound.

2.4 Equivalence between PAC-Bayes and Local Entropy

The most remarkable contribution of the article is proving that the Entropy-SGD objective is a surrogate for a PAC-Bayes bound. If we denote the empirical risk over the dataset S as $\hat{R}_S(w) = \frac{1}{m}\sum_{i=1}^{m} l(w, z_i)$, the "Local Entropy" $f_\gamma(w)$ is defined as:

$$f_\gamma(w) = -\log\left(G_\gamma * e^{-\hat{R}_S(w)}\right) = -\log\int_{\mathbb{R}^p} e^{-\hat{R}_S(w')} e^{-\frac{\gamma}{2}|w-w'|^2} dw'$$

Dziugaite and Roy demonstrate that with a prior P as a Gaussian centered at w (i.e., $\mathcal{N}(w, \gamma^{-1}I)$) and a posterior Q as the Gibbs distribution $e^{-\hat{R}_S(w')}$, minimizing Local Entropy is mathematically equivalent to minimizing the empirical risk and the KL-divergence term found in the PAC-Bayes bound:

$$f_\gamma(w) \propto \inf_Q \left(\mathbb{E}_{w'\sim Q}[\hat{R}_S(w')] + \mathrm{KL}(Q \,\|\, \mathcal{N}(w, \gamma^{-1}I))\right)$$

This identity provides a profound theoretical justification: since $\mathbb{E}_{w'\sim Q}[\hat{R}_S(w')] = \hat{R}_S(Q)$, the algorithm performs principled variational inference to directly minimize a generalization bound. The width of the valley is captured by the KL-divergence, where wider valleys allow for a posterior Q that is less "costly" in terms of information relative to the prior.

3 Implementation

Since there were no known Python implementations of Entropy-SGLD available, we had to rebuild the algorithm from scratch. During this process, we encountered several technical challenges.

3.1 General Insight

A primary challenge encountered during the implementation phase was the prohibitive computational overhead associated with the original algorithmic framework. Under the hyperparameter configuration originally proposed by the authors, the model exhibited negligible convergence even after long time execution (an hour just for a bit of progress), as the underlying optimization process inherently requires several thousands of epochs to reach stability. Consequently, we conducted an hyperparameter tuning phase aimed at accelerating the learning dynamics to fit within reasonable temporal constraints. This optimization involved a increase in the different learning rates, coupled with a systematic reduction of the injected noise. Specifically, by increasing the value of the parameter β, we reduced the noises then facilitates a more direct and rapid trajectory toward convergence.

However, this gain in computational efficiency introduced a trade-off regarding the theoretical robustness of the model. Since the precision and tightness of the PAC-Bayes bound are intrinsically dependent on the noise level and the β coefficient, our modifications resulted in a looser bound. Ultimately, while these adjustments allowed for empirical convergence within a manageable timeframe, they degraded the overall precision of the generalization guarantees provided by the PAC-Bayesian framework.

Beyond the challenges of achieving convergence, significant obstacles were encountered during the empirical evaluation of the PAC-Bayesian generalization bounds. Specifically, the methodology prescribed in the original study for estimating the KL divergence proved to be numerically unstable in our experimental setting. Despite rigorous implementation, the divergence values frequently drifted into negative values. While we conducted extensive diagnostic tests in parallel with our hyperparameter search to rectify these inaccuracies, the estimator remained persistently unreliable.

Consequently, given the strict temporal constraints of the project and the priority placed on achieving stable model convergence, we elected to pivot away from the formal calculation of these bounds.

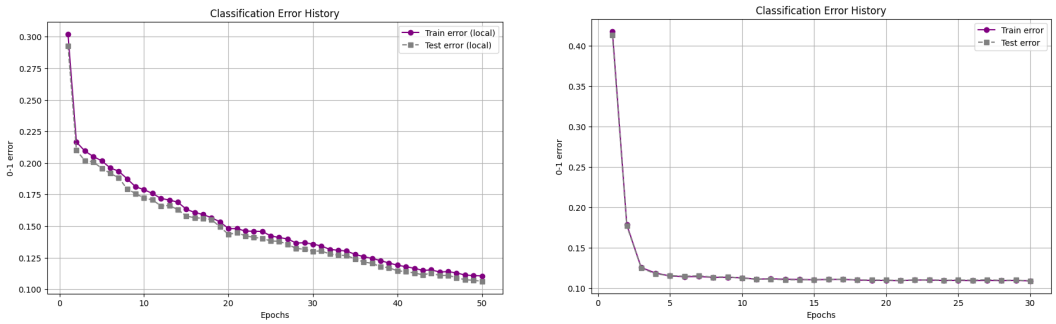
Furthermore, it is important to emphasize that under our specific hyperparameter configuration (notably β) we could have anticipated that the resulting PAC-Bayesian bounds would be excessively loose, ultimately providing minimal theoretical insight or practical utility for certifying the model's generalization performance.

We used the FC600 model as suggested to show the entropy-SGLD : three fully connected layers with 600 neurons and Relu activation.

3.2 Experiment and Results

We evaluated our implementation through three distinct experimental setups to validate the convergence, robustness, and generalization capabilities of the Entropy-SGLD algorithm:

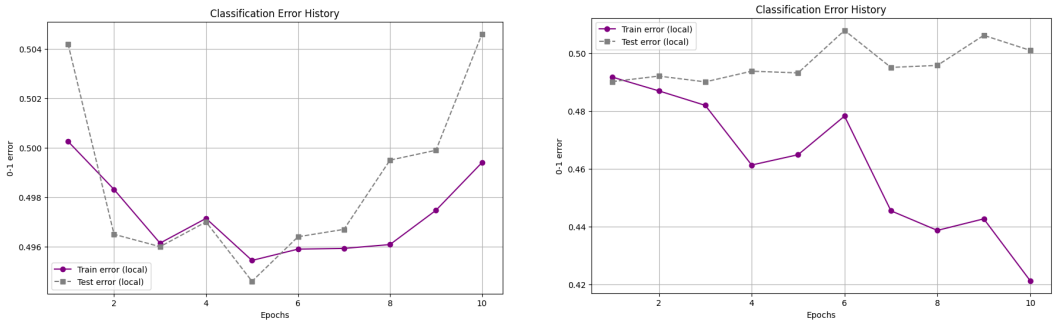
- **Standard Performance on Binary MNIST:** We first applied the model to a binarized MNIST dataset (same as in the original paper), where digits were partitioned into two classes (labels 0–4 vs. 5–9). The model showed consistent empirical convergence. Throughout the 50 training epochs, the test error tracked closely with the training error, frequently remaining slightly below it, which suggests that the algorithm maintains a strong inductive bias towards generalization.
- **Overfitting Stress Test (Random Labels):** To analyze the algorithm's resistance to noise memorization, we conducted a test using the same binary MNIST images but with randomly assigned labels. We compared Entropy-SGLD against a standard SGD baseline. While SGD rapidly overfitted by memorizing the random noise, Entropy-SGLD maintained significantly higher error rates. This confirms that the local entropy objective effectively prevents the model from collapsing into the "sharp" minima associated with noise, acting as a robust implicit regularizer.
- **Generalization to Fashion MNIST:** Finally, we tested the portability of our setup using the Fashion MNIST dataset (28×28 images of clothing and accessories). We applied the same binarization logic (grouping the first five classes and the remaining five) and maintained our previous hyperparameter configuration. The results demonstrated that the model successfully generalizes to more complex image manifolds beyond simple digits, showing a sharp initial decrease in error and stable learning behavior.



(a) MNIST (Entropy-SGLD)

(b) FashionMNIST (Entropy-SGLD)

Figure 1: Classification error history on true labels. Both MNIST (a) and FashionMNIST (b) show excellent generalization with Entropy-SGLD, as training and test errors overlap.



(a) Random Labels (Entropy-SGLD)

(b) Random Labels (SGD)

Figure 2: Comparison of optimization methods on MNIST with random labels. (a) Entropy-SGLD avoids memorization, maintaining an error rate near chance level. (b) Standard SGD exhibits severe overfitting, memorizing the training set while test error remains high.

4 Analysis and conclusion

The paper's primary strength lies in its rigorous theoretical framework: the authors demonstrate that maximizing local entropy equates to optimizing the prior of a PAC-Bayes bound on a Gibbs classifier. This insight exposes a flaw in standard Entropy-SGD, using a data-dependent prior invalidates the bound, and introduces Entropy-SGLD as a compelling alternative. By leveraging differential privacy via SGLD, this method successfully constructs valid, nonvacuous PAC-Bayes bounds that provide a formal certificate of generalization even in overparametrized settings.

However, the proposed methodology presents several notable weaknesses. On a theoretical level, the validity of the bounds relies on the optimistic and hard-to-verify assumption that the SGLD process has reached stationarity. Furthermore, the paper contains technical inaccuracies, such as a sign error in the final weight update step of Algorithm 1 (line 9). From a practical standpoint, the absence of an official Python implementation proved to be a major hurdle. Rebuilding the algorithm from scratch highlighted severe computational bottlenecks: Entropy-SGLD requires excessively long training times and learns much slower than standard methods. Ultimately, while the algorithm provides rigorous generalization certificates, its test performance remains far from state-of-the-art , illustrating a fundamental trade-off between learning speed, model accuracy, and theoretical guarantees.

Our investigation reveals that while Entropy-SGLD offers a mathematically elegant solution to the generalization bound, its current form acts more as a theoretical benchmark than a practical replacement for standard optimizers. Moving forward, the true challenge lies in refining these privacy-preserving mechanisms to a point where the cost of a formal generalization certificate no longer necessitates a significant compromise in competitive model accuracy.

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